

Mohd Moin Khan

M.Tech Robert Bosch Centre for Cyber Physical Systems



ACADEMIC PROFILE

M.Tech	7.6	Indian Institute of Science, Bangalore	2026
B.Tech	8.88	Rajasthan Technical University, Kota	2023

SKILLS

- **Programming:** C, C++, Python, Micropython, Shell Scripting
- **Tools:** Git, Docker, Label Studio, Jupyter, VS Code, Linux, MATLAB, LaTeX
- **ML/DL Libraries:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, LangChain, OpenCV, vLLM, Pandas, NumPy
- **Core AI/ML Concepts:** - **LLMs & NLP:** Transformers, RAG, Fine-tuning, Seq2Seq, Knowledge Distillation - **Deep Learning:** CNNs, ResNet, GANs, Autoencoders, Attention Mechanism, Model Pruning & Quantization, Image Segmentation - **Machine Learning:** Supervised & Unsupervised Learning, Ensemble Methods(Bagging, Boosting, Stacking), Imbalanced Data Handling(SMOTE, Class Weights), Feature Engineering, Dimensionality Reduction(PCA, SVD, t-SNE), Model Evaluation & Optimization
- **Robotics & Embedded Systems:** ROS2, Gazebo, MoveIt2, Arduino, Raspberry Pi, Nicla, Jetson Nano, NRF52, OpenMV; experience with RTOS (FreeRTOS, RTX5, ARM MBED OS) and interfaces (UART, SPI, I²C, CAN, PWM, ADC, BLE, Wi-Fi)
- **Edge AI & Tools:** TensorFlow Lite Micro, Edge Impulse, ONNX Runtime, MJPEG Streaming, Git, Docker, VS Code, Linux, MATLAB

INTERNSHIPS

3 Months

Siemens	Research Intern	May'25-Aug'25
• Deep Learning-Driven PCB Segmentation and Camera-to-Robot Calibration – Built a PCB segmentation pipeline using U²-Net and SAM + Label Studio to accelerate and improve annotation. – Implemented eye-to-hand calibration of Intel RealSense D435 with Doosan A0509 manipulator using ROS 2 & MoveIt 2, applying multiple solvers for precise robot-camera alignment.		
Air India	Technical Intern	Jun'22-Jul'22
• – Worked on aircraft maintenance and inspection procedures. • – Gained hands-on experience with aeronautical systems and safety protocols.		

PROJECTS

Portable AI-Based Sign Language Translator	Project Lead	Feb'25-May'25
• Built a real-time portable ISL translator on Arduino Nicla Vision using quantized INT8 MobileNetV2 CNN (<1 MB), created and augmented a 3,000+ image dataset to train lightweight models for resource-limited deployment.		
Image Captioning using Deep Learning		Jan'25-Apr'25
• Built CNN-LSTM and deep fusion models (InceptionV3 encoder + LSTM decoder) to generate captions from images, trained/evaluated on Flickr8k with BLEU, METEOR, and ROUGE-L metrics. Demonstrated that global feature fusion achieves competitive performance vs. attention-based models while reducing computational overhead.		
Implemented Transformer from Scratch: Neural Machine Translation (English to Hindi)		Jan'25-Mar'25
• Built a Transformer-based NMT model from scratch using PyTorch, implementing custom tokenization, attention mechanisms, and sequence optimization, gaining expertise in deep learning, NLP, and model optimization.		
Exploratory Data Analysis for Greater Melbourne Suburbs		Oct'24-Nov'24
• Analyzed demographic, healthcare, and land usage data using MDS, created 2D scatter plots, and provided urban planning insights. Skills: MDS, Data Visualization, Clustering, Statistical Analysis.		
Stack Bot		Aug'24-Nov'24
• Developed a robotic system with precise motion control using real-time feedback, PWM-based motor control with a PID Controller, rotary encoders, H-bridge drivers, and wireless radio control for remote communication via microbits.		
MPC Controller for Stoch - Quadraped Robot		Sep'24-Dec'24
• Developed an MPC-based controller in Python and C++ for trajectory optimization, integrating joint constraints, stability margins, ROS 2, and Gazebo simulation to achieve stable and efficient locomotion across terrains.		
Building Classification for Disaster Risk Management		Aug'24-Sep'24
• Developed an automated building classification system using CNNs to categorize Google Street View images into five types, optimizing models with ResNet and EfficientNets, processing 2,516 images, addressing class imbalance, and improving accuracy for disaster risk assessment.		
Blended Wing Body Aircraft		Aug'22-Jun'23
• Led the design, simulation, and analysis of a BWB aircraft, optimizing fuel efficiency and performance, while managing a team and achieving top project status.		

COURSES

Machine Learning for Cyber Physical Systems (PG)
Machine learning for Signal Processing (PG)
Agentic and Generative AI (PG)
Edge AI (PG)

Design of Cyber Physical Systems (PG)
Real Time Embedded Systems (PG)
Robotics Perception (PG)
Foundation of Robotics (PG)
Mathematical Techniques for Robotics and Autonomous Systems (PG)
Applied Linear and Non-linear Control (PG)
Operation Research for Mobility Management (PG)

POSITION OF RESPONSIBILITY

- Teaching Assistant of Data Science in Practice. Delivered tutorials for **90+ PG** students on ML and DL fundamentals including **Neural Networks, CNNs, Object Detection, Segmentation, Decision Trees, Regression, and EDA**, enhancing conceptual clarity and practical coding skills.
- Teaching Assistant of Design of Cyber-Physical-Systems. Conducted tutorials for **30+ PG** students on **GPIO drivers, encoders, radio control, motor interfacing, and PID control**, integrating control theory with embedded system design.
- Department Representative of RBCCPS IISc, Bangalore. Organized the **Freshers' Event 2025** and coordinated academic, research, and outreach initiatives.
- Flying Mustangs Aeromodelling Club - Coordinator: Aug 2021 – July 2023: Mentored 80+ students in aeromodelling, IoT, and robotics; organized a water rocket competition with 700+ participants.

ACHIEVEMENTS

- Secured AIR 130 in GATE 2024, AIR 676 in GATE 2023, and AIR 520 in GATE 2022.
- Qualified for the Boeing National Aeromodelling Competition at IIT Bombay 2022.
- Winner of the National level Water Rocket in THAR 2020.
- Runner-up in the National level Robo-War competition in Plinth 2020.
- Best Student Award in 12th standard of Army Public School Alwar 2018.